Abhinav Kesarwani

Generative AI | LLM Engineering | Machine Learning | Software Engineering
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Education

Gati Shakti Vishwavidyalaya **Vadodara, Gujarat**
B.Tech. in Artificial Intelligence & Data Science | CGPA: 8.91/10 *Aug 2023 – Aug 2027*

Technical Skills

Programming Languages: Python, C++, SQL
Machine Learning: Machine Learning, Deep Learning, PyTorch, Scikit-learn, Model Training, Explainable AI
Generative AI: RAG, LLM Fine-tuning (LoRA), Prompt Engineering, Transformer Models, Semantic Search
Computer Vision: Computer Vision, Object Detection, OCR, Image Processing
LLM & Retrieval: FAISS, BM25, Sentence Transformers, Cross-Encoder Reranking
Frameworks & Backend: Hugging Face Transformers, LangChain, LangGraph, FastAPI, Flask, Streamlit
Data Science & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI
Engineering & MLOps: ML Pipelines, Data Validation, GitHub Actions, CI/CD, Docker, REST APIs, Microservices
Cloud, Databases & Tools: AWS, EC2, Kubernetes, GCP, MySQL, PostgreSQL, Redis, Linux, Power Automate, Git

Professional Experience

Siemens **Pune, Maharashtra**
Technical Intern *May 2026 – August 2026*

- Developed and delivered **5 Python-based engineering automation projects** across software assurance, test management, data transformation, report generation, and quiz processing; engineered **RailAssure**, a FastAPI/React static analysis platform supporting **10 language families** with security, reliability, and compliance analysis.
- Automated **5 RQM maintenance tasks** and spreadsheet/report workflows, achieving an estimated **15% cumulative time saving** over corresponding manual processes.

Delhi Metro Rail Corporation (DMRC) **New Delhi**
IT Intern *May 2025 – Jun 2025*

- Developed and deployed an **Indoor Navigation System** for the DMRC Academy, supporting **100+ daily visitors** with mobile browser-based, QR-enabled navigation across indoor and multi-floor environments.
- Automated **trip-chart generation** to reduce repetitive manual processing and streamline operational workflow for transportation planning.

Projects

SecureDocAI — Privacy-Preserving Offline RAG Framework Jan 2025 – Present

- Built a fully offline **RAG** framework using FAISS, BM25, Sentence Transformers, Cross-Encoder reranking, OCR, and local SLMs for question answering over **35K+ indexed chunks** from **25+ documents**.
- Engineered hybrid dense-lexical retrieval and benchmarked **7 ablation configurations** using **125 manually verified questions** and **15+ evaluation metrics** covering retrieval, answer quality, latency, and faithfulness.
- Proposed an **Evidence-Based Confidence Score (EBCS)** to improve transparency and reliability of offline LLM-generated responses.

AI Interview Coach — Real-Time Voice AI Interview Platform GitHub

- Built a real-time **speech-to-speech AI interviewer** using **Pipecat, FastAPI, WebRTC, Deepgram STT, Groq LLM, and Cartesia TTS** for low-latency voice-based technical interviews.
- Engineered **context-aware follow-up question generation** using job descriptions and optional resumes, enabling personalized multi-turn conversations with the AI interviewer.
- Developed an asynchronous **React + FastAPI** architecture for browser-based audio streaming, LLM inference, and real-time AI voice responses using WebRTC.

Research

- **Accident Analysis and Safety Improvements in Indian Railways** — Research paper accepted at **ICICST 2026**, to appear in **SSRN Proceedings (Elsevier)**.

Achievements

- Secured **2nd Place** in **two university-level hackathons** for AI-powered transportation and enterprise software solutions.
- Ranked **3rd** in the **HCL GUVI Scholarship Program**.
- Conducted **Power BI workshops** for **40+ Indian Air Force officers** on data visualization and decision-support analytics.

Leadership & Certifications

- **Leadership:** Placement Coordinator; Internship Coordinator; Marketing & PR Co-Coordinator, Business Club.
- **Certifications:** NVIDIA — Applications of AI for Predictive Maintenance; Fundamentals of Deep Learning; Computer Vision for Industrial Inspection; IBM — Artificial Intelligence Fundamentals.